LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Problem Statement: Municipal Solid Waste Route Optimizer

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Problem Statement

Municipal Solid Waste Route Optimizer

The detailed official problem-statement text was not included in the provided lab handout. The requirements below are prepared around the stated system title and should be checked against the instructor-provided problem statement before final submission.

1. Requirements Analysis

The Municipal Solid Waste Route Optimizer is intended to support municipal waste collection operations by managing waste collection points and vehicles and generating optimized collection routes for waste collection drivers.

2. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow the municipal administrator to add and manage waste collection points.	High	When valid collection point details are entered, the system shall store the collection point and display it in the collection point list.	Collection points are required for planning waste collection routes.	Clear and testable requirement.
FR-002	Functional	The system shall allow the municipal administrator to register and manage waste collection vehicles.	High	When valid vehicle details are entered, the system shall add the vehicle to the available vehicle list.	Vehicle information is required for route planning.	Vehicle capacity should be maintained.
FR-003	Functional	The system shall generate an optimized waste collection route based on registered collection points and available vehicles.	High	Given valid collection points and available vehicles, the system shall generate a route containing the required collection points.	Route optimization is the core functionality of the system.	Route generation should consider available vehicle constraints.
FR-004	Functional	The system shall allow waste collection drivers to view	High	When a route is assigned to a driver, the driver shall be able to	Drivers require route information to perform waste	Route information should be clearly displayed.

		their assigned collection route.		view the collection points and their sequence.	collection.	
FR-005	Functional	The system shall allow waste collection drivers to update the collection status of assigned collection points.	High	When a driver marks a collection point as collected, the system shall record the point as completed.	Status updates allow municipal authorities to monitor collection progress.	Status should be updated immediately.
NFR-001	Nonfunctional	The system shall generate an optimized collection route within 10 seconds for a route containing up to 100 collection points.	High	A performance test shall confirm that the route is generated within 10 seconds for 100 collection points.	Fast route generation ensures efficient route planning.	Performance requirement.
NFR-002	Nonfunctional	The system shall be available during municipal waste collection operating hours.	High	During operating hours, at least 99% of availability checks shall successfully access the system.	Waste collection operations depend on system availability.	Availability requirement.

3. Peer Critique and Revision

Each requirement was reviewed for clarity, testability and relevance. The descriptions use unambiguous 'shall' statements and the acceptance criteria provide measurable conditions for determining whether each requirement is satisfied.

4. Actors

Actor	Description
Municipal Administrator	Manages collection points, vehicles and route generation.
Waste Collection Driver	Views assigned routes and updates collection status.
GPS/Map Service	Provides location information required for route generation.

5. Use Cases

Use Case ID	Use Case	Description
UC-01	Manage Collection Points	Add and manage waste collection points.
UC-02	Manage Vehicles	Register and manage collection vehicles.
UC-03	Generate Optimized Route	Generate an optimized route for waste collection.
UC-04	View Assigned Route	Allow drivers to view their assigned route.
UC-05	Update Collection Status	Record the collection status of collection points.
UC-06	Monitor Collection Progress	Monitor the progress of waste collection.
UC-07	Provide Location Data	Provide location information for route generation.

6. Use-Case Relationships

Include Relationship:

UC-03 Generate Optimized Route <<include>> **UC-07 Provide Location Data**

Reason: The route generation process requires location information to calculate an optimized collection route.

The UML use-case diagram represents the interactions between the actors and the Municipal Solid Waste Route Optimizer system. The diagram should include the three identified actors, the use cases, associations and the <<include>> relationship between route generation and location data.

7. Use-Case Flow

7.1 Use Case Information

Use Case ID: UC-03

Use Case Name: Generate Optimized Route

Primary Actor: Municipal Administrator

Supporting Actor: GPS/Map Service

7.2 Preconditions

1. The administrator is logged into the system.
2. Waste collection points have been registered.
3. At least one suitable collection vehicle is available.
4. Required location information is available.

7.3 Main Success Scenario

5. The Municipal Administrator selects Generate Optimized Route.
6. The system retrieves the registered waste collection points.
7. The system retrieves the available collection vehicles and their capacities.
8. The system requests location information from the GPS/Map Service.
9. The GPS/Map Service provides the required location information.
10. The system analyzes the collection points and available vehicles.
11. The system calculates an optimized waste collection route.
12. The system displays the optimized route and collection sequence.
13. The administrator reviews and confirms the generated route.
14. The system saves the confirmed route.
15. The system makes the route available for assignment to a waste collection driver.
16. The use case ends successfully.

7.4 Alternate Flow — No Suitable Vehicle Available

17. The system determines that no suitable vehicle is available for the required collection.
18. The system displays a notification to the administrator.
19. The system asks the administrator to add or assign a suitable vehicle.
20. The administrator makes a suitable vehicle available.
21. The system attempts to generate the route again.
22. If a suitable vehicle is still unavailable, the system cancels route generation.
23. The system notifies the administrator that the route could not be generated.

7.5 Alternate Flow — GPS/Map Service Unavailable

24. The system requests location information from the GPS/Map Service.

25. The GPS/Map Service fails to respond.
26. The system displays an error message.
27. The system asks the administrator to retry the request.
28. The administrator selects the retry option.
29. The system requests the location information again.
30. If the service responds successfully, the main flow continues from Step 5.
31. If the service remains unavailable, the system cancels route generation and notifies the administrator.

7.6 Postconditions

Successful completion:

- An optimized collection route has been generated.
- The route has been saved.
- The route is available for assignment to a driver.

Unsuccessful completion:

- No route is generated or saved.
- The administrator is notified of the reason for failure.

8. Conclusion

The requirements for the Municipal Solid Waste Route Optimizer were identified and documented using functional and nonfunctional requirements. The identified actors and use cases were used to develop a UML use-case model, including an <<include>> relationship. The Generate Optimized Route use case was further specified using a main success scenario and alternate flows to represent normal and exceptional system behavior.

9. UML Use-Case Diagram

